Siyu Yuan

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RESEARCH INTEREST

Siyu Yuan（员司雨）is a third-year Ph.D. student at Fudan University in the School of Data Science. Her interested research topics are mostly around cognitive science with generative agents, including (but are not limited to):

- **Cognitive reasoning**, especially on exploring the cognitive reasoning abilities of generative agents, including analogical reasoning, theory of mind, and belief exploration of LLMs. The ultimate goal is to enhance the understanding of these agents about themselves and others, thereby enabling them to generate responses that align better with human cognition.
- **Strategic Planning**, especially on equipping generative agents with human-level planning capabilities revolving around tool invocation, case-based reasoning, and other planning tasks.
- **Knowledge Acquisition**, especially on excavating knowledge based on generative agents, including concept acquisition, script generation and analogy making. These aim to construct rich knowledge bases that can be effectively utilized.

EDUCATION

Ph.D. in Statistics. **Fudan University**
Knowledge Works Laboratory at Fudan University, Advisor: Prof. Deqing Yang. *2021-2026 (estimated)*

B.S. in Data Science. **Fudan University**
Knowledge Works Laboratory at Fudan University, Advisor: Prof. Deqing Yang. *2017-2021*

SELECTED PUBLICATIONS

[*: Equal Contribution]

[1] *EASYTOOL: Enhancing LLM-based Agents with Concise Tool Instruction*
Siyu Yuan*, Kaitao Song*, Jiangjie Chen, Xu Tan, Yongliang Shen, Ren Kan, Dongsheng Li, Deqing Yang
Long paper, Preprint, 2024.

[2] *AnalogyKB: Unlocking Analogical Reasoning of Language Models with A Million-scale Knowledge Base*
Siyu Yuan*, Jiangjie Chen*, Jiaqing Liang, Changzhi Sun, Yanghua Xiao, Deqing Yang
Long paper, Preprint, 2023.

[3] *Beneath Surface Similarity: Large Language Models Make Reasonable Scientific Analogies after Structure Abduction*
Siyu Yuan, Jiangjie Chen, Xuyang Ge, Yanghua Xiao, Deqing Yang
Long paper, in: **EMNLP 2023** (Findings).

[4] *Distilling Script Knowledge from Large Language Models for Constrained Language Planning*
Siyu Yuan, Jiangjie Chen, Ziquan Fu, Xuyang Ge, Soham Shah, Charles Robert Jankowski, Yanghua Xiao, Deqing Yang
Long paper, in: **ACL 2023**. Outstanding Paper Award (1%)

[5] *Causality-aware Concept Extraction based on Knowledge-guided Prompting*
Siyu Yuan, Deqing Yang, Jinxi Liu, Shuyu Tian, Jiaqing Liang, Yanghua Xiao, Rui Xie
Long paper, in: **ACL 2023**.

[6] *Generative Entity Typing with Curriculum Learning*
Siyu Yuan, Deqing Yang, Jiaqing Liang, Zhixu Li, Jinxi Liu, Jingyue Huang, Yanghua Xiao
Long paper, in: **EMNLP 2022**.

[7] *Large-Scale Multi-granular Concept Extraction Based on Machine Reading Comprehension*
Siyu Yuan, Deqing Yang, Jiaqing Liang, Jilun Sun, Jingyue Huang, Kaiyan Cao, Yanghua Xiao, Rui Xie.
Long paper, in: **ISWC 2021**.

COMPLETE PUBLICATIONS

[*: Equal Contribution]

[8] *Deciphering Figurative Meanings: IdiomKB's Role in Optimizing Idiomatic Translation on Language Models*
Shuang Li, Jiangjie Chen, **Siyu Yuan**, Xinyi Wu, Hao Yang, Shimin Tao, Yanghua Xiao
Long paper, in: **AAAI 2024**.

[9] *Put Your Money Where Your Mouth Is: Evaluating Strategic Planning and Execution of LLM Agents in an Auction Arena*
Jiangjie Chen, **Siyu Yuan**, Rong Ye, Bodhisattwa Prasad Majumder, Kyle Richardson
Long paper, Preprint, 2023.

[10] *TaskBench: Benchmarking Large Language Models for Task Automation*
Yongliang Shen*, Kaitao Song*, Xu Tan, Wenqi Zhang, Kan Ren, **Siyu Yuan**, Weiming Lu, Dongsheng Li, Yueting Zhuang
Long paper, Preprint, 2023.

[11] *Contextual Information and Commonsense Based Prompt for Emotion Recognition in Conversation*
Jingjie Yi, Deqing Yang, **Siyu Yuan**, Kaiyan Cao, Zhiyao Zhang and Yanghua Xiao
Long paper, in: **ECML PKDD 2022**.

AWARDS AND ACHIEVEMENTS

ACL 2023 Outstanding Paper Award *2023*
- Association of Computational Linguistics

Huatai Securities Scholarship *2023*
- Fudan University

Outstanding Graduate Student of Shanghai Colleges and University *2021*
- Fudan University

Outstanding Student pacemaker of Fudan University *2021*
- Fudan University

China National Scholarship *2018*
- Fudan University

EXPERIENCE

Microsoft Research Asia (Shanghai) *September 2023 - Present*

- Role: Research Intern
- Advisor: Dr. Kaitao Song and Dr. Kan Ren
- Topics: Autonomous Agents with Planning and Tool Use
- Project: Participate in the **JARVIS** project (Github Stars: 22.1K) [1] [10]

ByteDance AI-Lab (Shanghai) *January 2023 - May 2023*

- Role: Research Intern
- Advisor: Dr. Changzhi Sun and Dr. Jiaze Chen
- Topics: Evaluation and Improvement on the analogical reasoning abilities of LLMs [2].

Brain Technologies, NLP Research Intern (Remote) *June 2022 - September 2022*

- Role: Research Intern
- Advisor: Dr. Charles Jankowski
- Topics: Symbolic Knowledge Distillation for Language Planning and LLM Prompt Engineering [4].

Knowledge Works Lab at Fudan University (Shanghai) *March 2019 - Present*

- Role: Student Researcher
- Advisor: Prof. Yanghua Xiao and Dr. Jiangjie Chen
- Topics: Knowledge Acquisition, e.g, Analogy-making [3], Entity Typing [6], and Concept Extraction [7] [5].

Academic Service

- Program Committee/Reviewer of EMNLP (2022, 2023), APIN (2022, 2023), ACL(2023), ARR (2023, 2024), IEA/AIE (2023, 2024), ICLR TinyPapers (2024)